

Friday seminar on topology

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This is a low-dimensional topology student seminar. This seminar has no fixed theme, and it mainly consists of participants presenting their recent work or articles they have read. Our topics cover Floer homology, invariants of knots and links, mapping class groups of 4-manifolds, embedded arc calculus, and so on. The seminar is usually held on Fridays from 3 PM to 5 PM in Room 210, Building No. 19.

2026 Spring Schedule

Lecture	Date	Lecturer	Topics	Papers
1	Mar 20	Bowen Tang	Meridian-traceless $SU(2)$ -representations for link groups	[XZ23]
2	Mar 27	Bowen Tang	Minimal $I^\natural$ property for links	
3	Apr 3	Bowen Tang	Cosmetic surgery for $\pm 1/n$ -surgeries	[DLE24]
4	Apr 10	Qizheng You	Construction of Hatcher-Wagoner's 1st obstruction	
No seminar on Apr 17, Apr 24, May 1, May 8 due to conferences and the May Day holiday				
5	May 15	Qizheng You	Construction of Hatcher-Wagoner's 2nd obstruction	
6	May 22	Xiayu Tan	Hatcher-Wagoner invariant for barbell diffeomorphisms	[Tan26]
7	May 29	Xingpei Liu	Framed instanton dimension for surgery manifolds	[GE25]
8	Jun 5	Bowen Tang	Different versions of surgery exact triangles	
9	Jun 12	TBA	TBA	

References

- [DLE24] Aliakbar Daemi, Tye Lidman, and Mike Miller Eismeier, *Filtered instanton homology and cosmetic surgery*, arXiv: 2410.21248 (2024).
- [GE25] Sudipta Ghosh and Mike Miller Eismeier, *Framed instanton homology and frøyshev's invariant*, arXiv: 2511.18885 (2025).
- [Tan26] Xiayu Tan, *Cerf diagrams and hatcher-wagoner invariants for barbell maps*, arXiv: 2604.00939 (2026).
- [XZ23] Yi Xie and Boyu Zhang, *On meridian-traceless $SU(2)$ -representations of link groups*, Adv. Math. **418** (2023), Paper No. 108947, 48. MR4558814